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RESEARCH

Interpretable hybrid metaheuristic optimization with iteration-level behavior analysis for clinical feature selection in heart disease prediction

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Abstract

Cardiovascular disease prediction demands accurate models alongside feature selection processes that are transparent, reproducible, and diagnostically interpretable. Existing wrapper-based metaheuristic selectors operate as black boxes, reporting only final accuracy without revealing how features are selected or whether subsets remain consistent across data samples. We propose WaOA-MRFO, an adaptive hybrid optimizer that combines Walrus Optimization Algorithm exploration with Manta Ray Foraging Optimization exploitation through stagnation-aware diversity control. A structured iteration-level diagnostic suite monitors population diversity, convergence behaviour, and feature selection stability, making the search dynamics directly observable rather than opaque. Under 5×5 nested cross-validation on the Cleveland heart disease cohort ($n = 261$), WaOA-MRFO achieves an AUC of 0.843 ± 0.056 — statistically equivalent to the all-features baseline (AUC $= 0.847$, $p = 0.619$) — while reducing the feature set by 61%. The method matches LASSO and recursive feature elimination on identical splits and significantly outperforms uninformed selection at matched dimensionality ($\Delta\text{AUC} = +0.091$, $p < 0.001$). A stable four-feature core (*exang*, *ca*, *thalch*, *oldpeak*) emerges in more than 70% of folds, while the top two features are selected in every fold, indicating reproducible signal extraction. EvoMapX operator attribution analysis identifies the WaOA spiral operator as the primary convergence driver, accounting for 66.7% of improvement events, while the Population Evolution Graph reveals the progressive stabilisation of the informative feature core across iterations. Fixed-mask transfer evaluation on an independent three-site cohort ($n = 616$) yields AUC $= 0.811 \pm 0.041$, confirming cross-institutional transferability of the stable-tier features. Ablation analysis identifies bit-flip mutation as the sole statistically significant contributor to subset consistency ($p = 0.028$, Cohen's $d = 0.531$. These results establish that competitive predictive performance with feature sets reduced by 61% is achievable alongside transparency, cross-institutional transferability, and process-level interpretability.

Keywords WaOA-MRFO · Hybrid optimization · Feature selection · Interpretable machine learning · Heart disease prediction

1 Introduction

Cardiovascular diseases (CVD) remain a leading cause of global mortality, accounting for an estimated 17.9 million deaths annually (Organization 2019); early and accurate risk stratification remains a cornerstone of preventive care (Selvam et al. 2025). The development of data-driven clinical decision-support systems typically relies on high-dimensional feature sets encompassing demographic, laboratory, and physiological variables (Tuppad and Patil 2022). These datasets are characterized by statistical correlations, redundancy, and noise, making feature selection (FS) not merely a preprocessing step but a clinical necessity for model interpretability, operational efficiency, and physiological insight (Sahu et al. 2025; Sakib and Corresponding 2025; Maruotto et al. 2025). A selection process that yields unstable or non-reproducible feature subsets undermines clinical trust and obstructs the identification of reliable biomarkers (Wong et al. 2025).

The methodological evolution of FS reflects an ongoing tension between performance and stability (Khaire and Dhanalakshmi 2022). Filter methods such as mutual information and correlation-based filters prioritize computational efficiency but assume feature independence, ignoring critical

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interactions in cardiovascular physiology (Noroozi et al. 2023; Yu et al. 2026). Wrapper methods address this limitation by embedding selection within classifier performance evaluation, capturing feature dependencies at the cost of subset instability across runs due to stochastic initialization (Tye et al. 2025; Khaire and Dhanalakshmi 2022). Embedded methods, including LASSO and tree-based importance measures, integrate selection with model training but bind results to a specific algorithm's inductive bias, limiting generalizability (Modeling et al. 2025). Across all three paradigms, evaluation has focused almost exclusively on terminal predictive metrics, leaving the dynamics and consistency of the selection process itself largely unexamined.

To navigate the combinatorial complexity of wrapper-based selection, evolutionary and swarm-based metaheuristics, including Genetic Algorithms (GA) (Naskar et al. 2025), Particle Swarm Optimization (PSO) (Liu et al. 2025), Walrus Optimization Algorithm (WaOA) (Trojovský and Dehghani 2023; Adamu et al. 2026), and Manta Ray Foraging Optimization (MRFO) (Adamu et al. 2025), have been widely adopted for their global search capability. However, their application in clinical FS reveals significant limitations: they are frequently deployed with static control parameters ill-suited to correlated clinical data (Rizk et al. 2025), and their hybridization logic is typically heuristic, relying on fixed schedules rather than runtime behavioral indicators (Xue et al. 2025). Consequently, these optimizers function as black boxes: studies report final subsets and accuracy but omit analysis of search trajectories, population diversity, and iteration-level feature stability (Chen et al. 2026).

A synthesis of these limitations reveals a core methodological gap: the disconnect between advanced optimization strategies and the clinical need for a stable and interpretable selection process. This gap manifests in three specific deficiencies:

1. **Non-Adaptive Hybridization:** Existing hybrids lack mechanisms for behavior-aware adaptation in which the integration of exploratory and exploitative search is governed by measurable runtime metrics such as population diversity and convergence rate (Chen et al. 2026).
2. **Black-Box Optimization Dynamics:** The field lacks iteration-level algorithmic behavior analysis. Convergence is reported post hoc, while diagnostics on feature inclusion frequency, subset transition patterns, and exploration–exploitation phases are largely omitted (Suman et al. 2026).
3. **Process-Level Interpretability:** Interpretability research in healthcare machine learning (ML) focuses primarily on model-level explanations including Shapley Additive Explanations(SHAP) and LIME (Abed-Alguni 2026), leaving the upstream optimization process that determines feature inclusion opaque and insufficiently examined.

This study addresses these gaps by proposing a process-transparent hybrid optimization framework for clinical FS. Unlike existing approaches that confine interpretability to post-hoc model explanations, the proposed framework exposes the internal dynamics of the optimization process through iteration-level analysis of FS behavior, as specified in Algorithm 1. The framework is evaluated on heart disease prediction, with the following contributions:

- We propose WaOA-MRFO, a population-based hybrid optimizer integrating WaOA exploration operators and MRFO exploitation operators through stagnation-aware diversity control and a diversity-sensitive adaptive weight. Observable stagnation triggers initiate diversity injection, enabling transparent regulation of convergence dynamics.
- A process-level interpretability diagnostic suite that formalizes optimization transparency by quantifying FS stability, population diversity evolution, and phase-transition behavior across iterative search trajectories, providing iteration-level insight into the selection process.
- A comprehensive evaluation framework validating the proposed approach across predictive performance, subset reproducibility, optimization transparency, and clinical plausibility of selected features, establishing a stability-aware benchmark for FS in clinical decision-support systems.

The remainder of this paper is organized as follows. Section 2 presents the literature review. Section 3 details the proposed methodology. Section 4 reports experimental results. Section 5 discusses findings, limitations, and future research directions.

2 Literature review

FS in heart disease prediction has progressed from statistical ranking techniques to population-based search strategies (Sowmiya et al. 2025). In cardiovascular decision-support systems, the objective extends beyond predictive accuracy to include reliable identification of physiologically meaningful markers (Netala et al. 2025). An unstable feature subset undermines clinical trust, complicates biomarker interpretation, and obstructs longitudinal validation of risk indicators. When repeated runs of a selection algorithm produce different subsets despite similar accuracy, any claimed

transparency becomes superficial (Sanchez-Martinez et al. 2021).

Filter-based approaches ranked features using mutual information, correlation coefficients, and statistical significance tests. These methods reduce dimensionality efficiently but assume variable independence, neglecting higher-order interactions common in cardiovascular physiology where correlated laboratory measures may redundantly retain weakly predictive features while discarding complementary ones (Rehman et al. 2025; Yu et al. 2026).

Wrapper methods addressed this limitation by embedding feature evaluation within classifier performance metrics (Tye et al. 2025; Khaire and Dhanalakshmi 2022), capturing nonlinear dependencies through predictive modeling. However, interaction awareness introduces stochastic variability: minor perturbations in initialization or data sampling can yield divergent feature subsets even when overall accuracy remains consistent. This instability limits reproducibility and challenges clinical interpretability.

Embedded methods, including regularization frameworks and tree-based ensembles, integrate the selection mechanism within model training. Although computationally efficient, these approaches entangle subset determination with model-specific inductive biases; a feature deemed important by one classifier may not persist across different architectures (Modeling et al. 2025). Across filter, wrapper, and embedded paradigms, stability analysis and subset consistency remain secondary to predictive performance.

The combinatorial complexity of wrapper-based selection motivated adoption of evolutionary and swarm-based metaheuristics. GA (Naskar et al. 2025) and Differential Evolution explore discrete spaces through mutation and recombination, while PSO (Liu et al. 2025), WaOA (Urundai Meeran 2025; Han et al. 2024), and MRFO (Zhao et al. 2019; Adamu et al. 2024) leverage collective search behaviors. These frameworks improve global exploration and mitigate entrapment in local optima. Most implementations rely on static parameter schedules that presume a homogeneous search landscape, which is rarely valid for clinical datasets characterized by clustered redundancy and sparse informative signals (Rizk et al. 2025).

Hybrid metaheuristics (Chen et al. 2025) were developed to balance exploration and exploitation by combining complementary optimizers. Advances in adaptive behavior within single metaheuristics through self-adaptive mechanisms and game-theoretic learning have been reported (Cuevas et al. 2024), but these are confined to intra-algorithm adaptation and do not address orchestration of multiple distinct algorithms. In practice, hybridization is frequently heuristic rather than behavior-driven (Sharma et al. 2025): switching is based on fixed iteration thresholds or predefined probabilities rather than measurable runtime indicators (Essaadoui et al. 2025; Mahmud et al. 2022). Few studies employ quantifiable diagnostics such as population entropy, improvement rate decay, or subset overlap metrics to trigger adaptive transitions (Budiman et al. 2025; Murugesh et al. 2025). Without formal linkage between search behavior and inter-algorithm control, hybrid frameworks risk functioning as layered heuristics rather than responsive systems.

Interpretability research in clinical ML has concentrated on model-level explanations including SHAP and LIME (Mohammadi et al. 2026). These techniques elucidate how trained classifiers weight input features for specific predictions, but their validity presupposes stability of the upstream feature subset. If the selection process is unstable or opaque, explanation of model outputs offers incomplete transparency (Wang et al. 2025). The FS process, including search trajectory, feature inclusion frequency, and cross-run stability, remains insufficiently examined.

Process-level transparency is further constrained by conventional convergence reporting practices (Abed-Alguni 2026). Convergence is typically presented through aggregate objective trajectories without detailed interrogation of iteration-level dynamics. Metrics capturing exploration intensity, exploitation phase transitions, and population diversity evolution are rarely analyzed in relation to subset stability (Chen et al. 2025; Rizk et al. 2025). Consequently, optimization-based selection often operates as a black box despite increasing algorithmic sophistication (Mohammadi et al. 2026).

In summary, the literature shows clear progress in global search strategies and hybrid optimization for FS. However, adaptive inter-algorithm control based on measurable runtime indicators remains limited, and interpretability has advanced primarily at the model level. The optimization process itself, including subset stability and search-phase behavior, is rarely examined in a systematic manner. These limitations restrict methodological transparency and leave process-level dynamics insufficiently understood in clinical FS frameworks.

3 Methods

A summary of all mathematical notation introduced in this section is provided in Table 12.

3.1 Problem formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ denote the dataset, where $\mathbf{x}_i \in \mathbb{R}^d$ is the i-th patient's clinical feature vector over d attributes and $y_i \in \{0, 1\}$ indicates angiographically confirmed coronary

artery disease. FS is cast as a constrained combinatorial optimization over the binary hypercube $\{0, 1\}^d$:

$$\mathbf{s}^* = \arg\max_{\mathbf{s}\in\{0,1\}^d} \left\{ f(\mathbf{s}) - \lambda \left(\frac{|\mathbf{s}|_0}{d} \right)^2 + \psi(\mathbf{s}) \right\} \quad (1)$$

where $\mathbf{s} = [s_1, \ldots, s_d]^\top$ is a binary feature mask with $s_j = 1$ indicating inclusion of feature j, $|\mathbf{s}|_0 = \sum_{j=1}^{d} s_j$ is the selected feature count, $f(\mathbf{s})$ is the predictive fitness function, $\lambda = 0.03$ is the quadratic parsimony penalty, and $\psi(\mathbf{s})$ is a domain-informed bonus function.

The fitness function evaluates each candidate subset via five-repeat Monte Carlo cross-validation (StratifiedShuffleSplit, validation fraction 0.25) on the training partition, using logistic regression as the fitness evaluator. In each split k, the best-performing regularization strength is selected independently:

$$f(\mathbf{s}) = \frac{1}{K} \sum_{k=1}^{K} \max_{C\in\mathcal{C}} \mathrm{AUC}_k(\mathbf{s}, C), \qquad K = 5, \quad \mathcal{C} = \{0.1,\ 1.0,\ 10.0\} \quad (2)$$

Per-split maximization over C yields an optimistic fitness estimate relative to a fixed-policy evaluator; this is acknowledged as a limitation in Section 5.6.

The bonus function encodes the clinical prior that cardiovascular disease assessment requires integration of multiple independent markers while excessive features introduce noise:

$$\psi(\mathbf{s}) = \begin{cases} 0.015 & \text{if } 5 \le |\mathbf{s}|_0 \le 7 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

3.2 Dataset and preprocessing

3.2.1 Study cohorts

Two complementary cohorts were used (Tables 1 and 2). The Core Evaluation Cohort (Cleveland Clinic Foundation, $n = 261$ after preprocessing) was sourced from the UCI ML Repository (Janosi Andras and Detrano 1989) and provides the primary benchmark for FS dynamics, algorithmic stability, and interpretability analysis across the full 13-dimensional feature space. The Generalization Cohort ($n = 920$, 10 features) aggregates data from four institutions (Table 2), providing a secondary assessment of feature subset transferability under heterogeneous demographic structures and a distinct class prevalence; generalization results are reported in Section 4.14.. The endpoint in both cohorts is angiographically confirmed coronary artery stenosis of at least 50%.

3.2.2 Preprocessing

The raw disease severity indicator (range 0–4) was binarized: severity 0 denotes absence of significant stenosis; values 1–4

Table 1 Clinical features, encoding, and cohort inclusion. *Core Only* features were excluded from the Generalization Cohort owing to institutional missingness exceeding 60% in at least one contributing site

Feature	Description	Type	Encoding	Cohort
age	Age (years)	Continuous	Numeric	Core & Gen
sex	Biological sex	Binary	1 = Male; 0 = Female	Core & Gen
cp	Chest pain type	Ordinal	0 = Typical; 1 = Atypical; 2 = Non-anginal; 3 = Asymptomatic	Core & Gen
trestbps	Resting BP (mm Hg)	Continuous	Numeric	Core & Gen
chol	Serum cholesterol (mg/dL)	Continuous	Numeric	Core & Gen
fbs	Fasting blood sugar > 120 mg/dL	Binary	1 = True; 0 = False	Core & Gen
restecg	Resting ECG results	Ordinal	0 = Normal; 1 = LV hypertrophy; 2 = ST-T abnormality	Core & Gen
thalch	Maximum heart rate achieved	Continuous	Numeric	Core & Gen
exang	Exercise-induced angina	Binary	1 = Yes; 0 = No	Core & Gen
oldpeak	ST depression (exercise vs. rest)	Continuous	Numeric	Core & Gen
slope	Slope of peak exercise ST	Ordinal	0 = Upsloping; 1 = Flat; 2 = Downsloping	Core Only
ca	Number of major vessels (0–3)	Discrete	Numeric	Core Only
thal	Thalassemia test result	Ordinal	1 = Normal; 2 = Fixed; 3 = Reversible defect	Core Only
target	Coronary stenosis ≥ 50%	Binary	1 = Disease; 0 = No disease	Core & Gen

Table 2 Cohort composition after preprocessing. The Core Cohort excluded 42 records with unresolvable missing values in `ca`, `thal`, or `slope`

Cohort	Source Institution	Original n	Excluded	Final n	Disease n (%)
Core (Cleveland)	Cleveland Clinic Foundation, USA	303	42	261	139 (53.3%)
	Hungarian Institute of Cardiology	294	0	294	138 (46.9%)
Generalization	University Hospital Zurich	123	0	123	48 (39.0%)
(Multi-centre)	VA Medical Center Long Beach, USA	200	0	200	96 (48.0%)
	Combined	**920**	**0**	**920**	**509 (55.3%)**

were collapsed to 1. Categorical variables were mapped to integer encoding per Table 1.

Missing value imputation was applied prior to train/test partitioning. Continuous features were filled with the dataset-level median; categorical features with the dataset-level mode. Forty-two Cleveland records were excluded prior to partitioning because missing values in *ca*, *thal*, or *slope* occurred within categories containing fewer than two observed instances, rendering mode imputation unreliable. For the Generalization Cohort, *slope*, *ca*, and *thal* were excluded entirely because at least one contributing site exhibited missingness exceeding 60%.

Continuous predictors were standardized to zero mean and unit variance. The scaler was fitted on the training partition of each split and applied to all downstream partitions, preventing leakage. Both cohorts were partitioned 80/20 (stratified) into a development set and a sealed holdout set; all nested cross-validation was conducted within the development set.

Both cohorts exhibit mild class imbalance (Core: 53.3%/46.7%; Generalization: 55.3%/44.7%). Stratified splitting was used throughout, and both the logistic regression fitness evaluator and the DNN classifier applied `class_weight='balanced'` to adjust the loss function by inverse class frequency.

3.3 Hybrid optimization framework

The proposed method synthesizes the WaOA (Trojovský and Dehghani 2023) and MRFO (Zhao et al. 2019) into a unified adaptive operator. Figure 1 provides an overview of the full pipeline.

3.3.1 Population initialization

The population $\mathbf{P}^{(0)} = \{\mathbf{s}_1^{(0)}, \ldots, \mathbf{s}_N^{(0)}\}$ of $N = 30$ solutions is initialized via stratified Bernoulli sampling across three sparsity strata, ensuring the initial search covers low-, moderate-, and high-dimensional subspaces:

$$s_{i,j}^{(0)} \sim \text{Bernoulli}(p_i), \qquad p_i = \begin{cases} \mathcal{U}(0.2,\ 0.4) & i \le N/3 \\ \mathcal{U}(0.4,\ 0.6) & N/3 < i \le 2N/3 \\ \mathcal{U}(0.6,\ 0.8) & i > 2N/3 \end{cases} \quad (4)$$

3.3.2 Continuous solution representation

Each solution is maintained as a continuous vector whose components are mapped through a sigmoid function $\sigma(z) = 1/(1+e^{-z})$ to produce selection probabilities, then thresholded: $\hat{s}_j = \mathbb{I}[\sigma(z_j) > 0.5]$. To prevent numerical overflow, the sigmoid input is clipped to $[-10,\ 10]$ and, before the logit transformation, the probabilities are clipped to $[\varepsilon,\ 1-\varepsilon]$ with $\varepsilon = 10^{-6}$. All hybrid updates and mutations operate on these continuous values; binarization is applied as the final step before fitness evaluation.

3.3.3 Phase-dependent hybrid update

Phase selection At each iteration t, the algorithm selects between exploration and exploitation based on population diversity $D(t)$ and normalized progress $\tau = t/T_{\max}$:

$$\text{Phase}(t) = \begin{cases} \text{Exploration} & \text{if } D(t) < 0.2 \text{ or } \tau < 0.3 \\ \text{Exploitation} & \text{otherwise} \end{cases} \quad (5)$$

where $D(t) = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{d}\|\mathbf{b}_i - \bar{\mathbf{b}}\|_1$ is the mean normalized Hamming distance from the population centroid $\bar{\mathbf{b}} = N^{-1}\sum_i \mathbf{b}_i$, with $\mathbf{b}_i = \mathbb{I}[\mathbf{s}_i > 0.5]$. The low-diversity branch ($D(t) < 0.2$) triggers exploration as a corrective response to premature convergence, reintroducing search breadth when the population clusters prematurely.

Adaptive hybrid weight A single diversity-dependent weight $\omega(t)$ governs the balance between the two exploitation branches. It is updated via exponential moving average with a diversity-sensitive target:

$$\omega(t+1) = \delta\,\omega(t) + (1-\delta)\,\omega_{\text{target}}(t), \qquad \omega_{\text{target}}(t) = \begin{cases} 0.7 & D(t) < 0.1 \\ 0.3 & \text{otherwise} \end{cases} \quad (6)$$

with $\delta = 0.70$ and $\omega(0) = 0.5$. For iterations $t \le 10$, ω is held at its initial value of 0.5; the adaptive update activates from iteration 11 onward. The decay $\delta = 0.70$ yields a time constant of approximately three iterations once active, enabling rapid response to diversity changes.

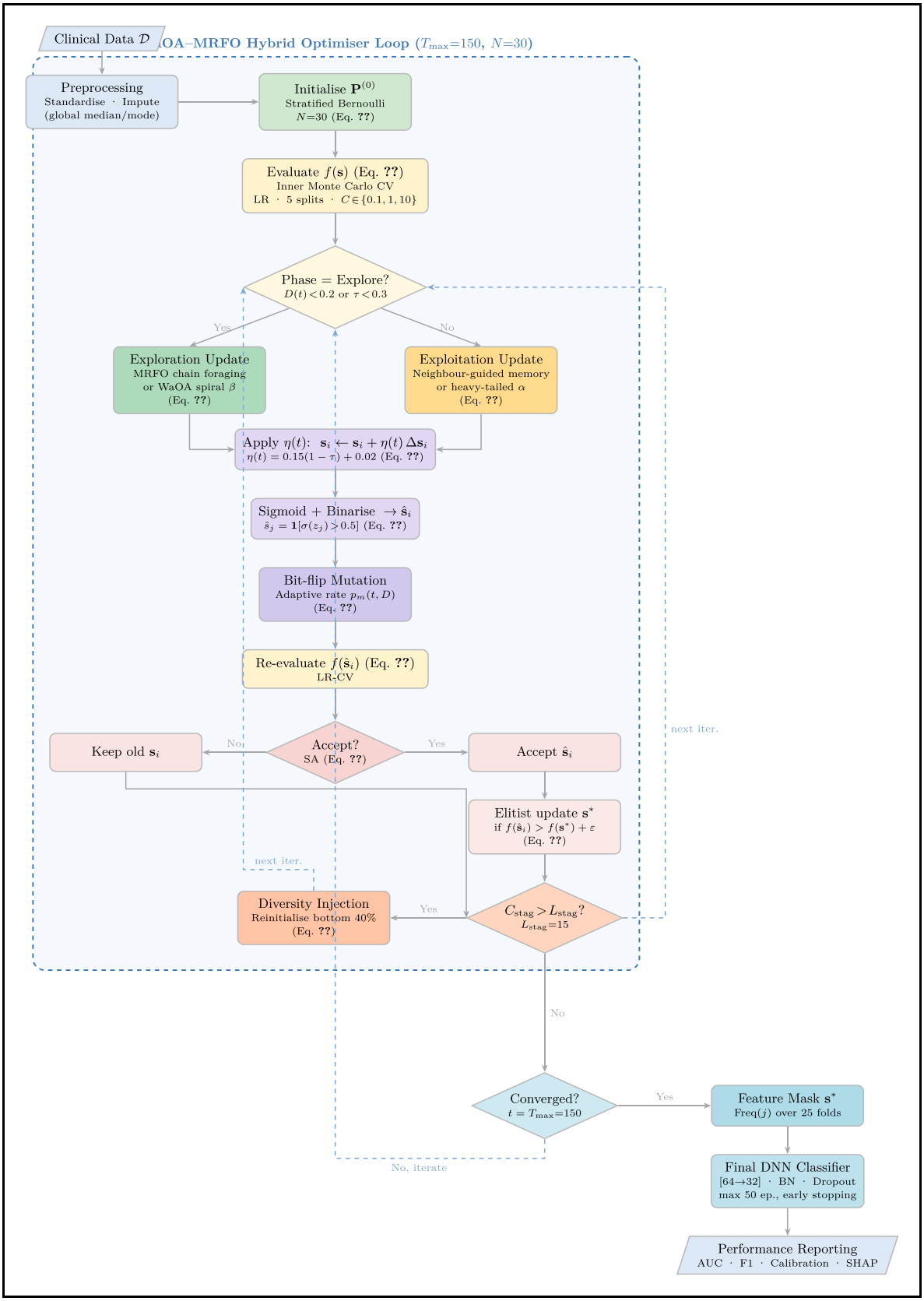


Fig. 1 WaOA-MRFO hybrid optimisation framework for heart disease FS.

Exploration update In exploration mode, the update is drawn from one of two branches with equal probability:

$$\Delta \mathbf{s}_i^{\text{explore}}(t) = \begin{cases} \boldsymbol{\xi} \odot (\mathbf{s}^* - I \cdot \mathbf{s}_i) & \text{prob. } 0.5 \text{ [MRFO chain foraging]} \\ \beta(t)\,(\mathbf{s}^* - \mathbf{s}_i) + 0.1\,\boldsymbol{\xi} & \text{prob. } 0.5 \text{ [WaOA spiral]} \end{cases} \tag{7}$$

where $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $I \sim \text{Uniform}\{1, 2, 3\}$, and $\beta(t) = 2e^{r(1-\tau)} \sin(2\pi r)$ with $r \sim \mathcal{U}(0, 1)$.

Exploitation update In exploitation mode, the weight $\omega(t)$ selects between neighbor-guided adaptive memory and a heavy-tailed perturbation:

$$\Delta \mathbf{s}_i^{\text{exploit}}(t) = \begin{cases} \mathbf{r} \odot (\tilde{\mathbf{s}}^* - \mathbf{s}_i) & \text{prob. } \omega(t) \\ \alpha(t)\,(\mathbf{s}^* - \mathbf{s}_i) & \text{prob. } 1 - \omega(t) \end{cases} \tag{8}$$

where $\mathbf{r} \sim \mathcal{U}(0, 1)^d$, $\tilde{\mathbf{s}}^* = \mathbf{s}^* + \boldsymbol{\xi} \cdot 0.1$ is a stochastically perturbed neighbor of the global best, and $\alpha(t) = 2r\sqrt{-\log(r + 10^{-7})}$ with $r \sim \mathcal{U}(0, 1)$ generates a moderately heavy-tailed step size.

Learning rate The step magnitude is scaled by a linearly decaying learning rate:

$$\mathbf{s}_i^{(t+1)} \leftarrow \mathbf{s}_i^{(t)} + \eta(t)\,\Delta \mathbf{s}_i^{(t)}, \qquad \eta(t) = 0.15(1-\tau) + 0.02 \tag{9}$$

yielding $\eta(0) = 0.17$ at the start and $\eta(T_{\max}) = 0.02$ at convergence.

3.3.4 Stochastic bit-flip mutation

After the hybrid update, a bit-flip mutation is applied to the binarized candidate to prevent premature convergence:

$$p_m(t, D) = \begin{cases} \max(0.12,\ p_{\text{base}}) & \tau > 0.7 \text{ and } D < 0.2 \\ 0.08 & D < 0.25 \\ 0.03 & \text{otherwise} \end{cases} \tag{10}$$

If the mutated candidate is identical to its parent, a forced flip at a uniformly random position guarantees population evolution.

3.3.5 Selection with simulated annealing

Acceptance at the population-member level follows a temperature-modulated rule. If the candidate fitness exceeds the current member fitness, acceptance is greedy. Otherwise, probabilistic acceptance applies:

$$\mathbf{s}_i^{(t+1)} = \begin{cases} \hat{\mathbf{s}}_i^{(t+1)} & \text{if } f(\hat{\mathbf{s}}_i^{(t+1)}) > f(\mathbf{s}_i^{(t)}) \\ \hat{\mathbf{s}}_i^{(t+1)} & \text{with prob. } \exp(\Delta f / T(t)) \text{ otherwise} \end{cases} \tag{11}$$

where $\Delta f < 0$ and $T(t) = 0.1(1 - \tau)$ decreases linearly from $T(0) = 0.1$ to $T(T_{\max}) = 0$. The global best solution is preserved elitistically and updated exclusively by strict improvement:

$$\mathbf{s}^* \leftarrow \mathbf{s}_i^{(t+1)} \quad \text{if} \quad f(\mathbf{s}_i^{(t+1)}) > f(\mathbf{s}^*) + 10^{-7} \tag{12}$$

The global best is therefore never replaced by a worse solution, regardless of the SA acceptance outcome.

3.3.6 Stagnation-triggered diversity injection

A stagnation counter $C_{\text{stag}}(t)$ tracks iterations since the last global-best improvement. When $C_{\text{stag}}(t) > L_{\text{stag}} = 15$, the bottom 40% of solutions by fitness are reinitialized:

$$\mathbf{s}_i^{(t)} \sim \text{Bernoulli}(p_{\text{reset}}), \quad p_{\text{reset}} \sim \mathcal{U}(0.2, 0.7), \quad \forall\, i \in \mathcal{I}_{\text{worst}} \tag{13}$$

where $\mathcal{I}_{\text{worst}} = \{i : \text{rank}(f(\mathbf{s}_i^{(t)})) \le \lfloor 0.4N \rfloor\}$. The stagnation counter resets to zero after each injection.

3.4 Final predictive classifier

The hybrid optimizer uses logistic regression for fitness evaluation, requiring up to $N \times T_{\max} \times K = 67{,}500$ model trainings (worst case); substituting a deep network at this stage would increase per-fold runtime by approximately 30-fold, making nested cross-validation computationally infeasible. The selected feature mask $\mathbf{s}^*$ is classifier-agnostic (Hancock et al. 2025); pilot experiments showed a deep network achieves 1–2% higher AUC on held-out data using the same selected features.

Following FS, a feedforward network is trained on $\mathbf{x}_{\text{sel}} \in \mathbb{R}^{|\mathbf{s}^*|_0}$:

$$\begin{aligned} \mathbf{h}^{(0)} &= \text{BN}(\mathbf{x}_{\text{sel}}) \\ \mathbf{h}^{(\ell)} &= \text{Dropout}\big(\text{BN}(\text{ReLU}(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)})),\ p{=}0.3\big), \quad \ell = 1, 2 \\ \hat{y} &= \sigma\left(\mathbf{w}^{(3)\top}\mathbf{h}^{(2)} + b^{(3)}\right) \end{aligned} \tag{14}$$

where $\mathbf{W}^{(1)} \in \mathbb{R}^{64 \times |\mathbf{s}^*|_0}$, $\mathbf{W}^{(2)} \in \mathbb{R}^{32 \times 64}$, and $\text{BN}(\cdot)$ denotes batch normalization. Weights are initialized with He normal initialization (He et al. 2015). Regularization combines

Dropout ($p = 0.3$) and L2 weight decay ($\lambda_{L2} = 0.001$) on all dense layers.

The network minimizes binary cross-entropy with L2 regularization:

$$\mathcal{L}(\boldsymbol{\Theta}) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right] + \lambda_{L2}\sum_{\ell=1}^{2}\|\mathbf{W}^{(\ell)}\|_F^2 \quad (15)$$

Training uses the Adam optimizer ($\eta_0 = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$). Within each outer CV fold, the outer training partition is further divided 80/20 (stratified) to produce a validation set for early stopping (patience 10 epochs) and learning rate reduction on plateau (factor 0.5, patience 5 epochs, minimum 10^{-6}). A maximum of 50 epochs is permitted; weights yielding the lowest validation loss are restored at the end of training. Batch size is 32, and `class_weight='balanced'` is applied throughout.

3.5 Computational complexity

Per-iteration cost decomposes as:

$$\mathcal{T}_{\text{iter}} = \underbrace{N \cdot K \cdot |\mathcal{C}| \cdot \mathcal{O}(nd)}_{\text{fitness evaluation}} + \underbrace{\mathcal{O}(Nd)}_{\text{hybrid operator}} + \underbrace{\mathcal{O}(Nd)}_{\text{diversity (centroid)}} \quad (16)$$

Total worst-case complexity is $\mathcal{O}(T_{\max} \cdot N \cdot K \cdot |\mathcal{C}| \cdot nd)$, yielding 67,500 fitness evaluations. Empirically, 9,626 evaluations were performed (14.3%) due to fitness caching via binary mask hash lookup. Table 3 places the hybrid method alongside all baselines.

3.6 Evaluation protocol

3.6.1 Nested cross-validation

Generalization performance was estimated via 5×5 nested cross-validation (RepeatedStratifiedKFold, 5 outer folds, 5 repeats, distinct random seeds), yielding 25 independent train–test splits from the development set ($n = 208$). In each outer fold, the optimizer runs on the outer training partition with five-repeat Monte Carlo inner cross-validation (StratifiedShuffleSplit, validation fraction 0.25) for fitness evaluation; the outer 20% is held out for performance reporting. The sealed holdout ($n = 53$) was evaluated once after all nested CV analysis was complete and is reported separately as a supplementary estimate. This nested structure prevents information leakage between FS and performance estimation (Abedin et al. 2026).

3.6.2 Baseline comparisons

Five baselines were evaluated under identical outer splits. The All-Feature baseline applies the DNN to all 13 features without selection. The Random-k baseline randomly samples $k \sim \text{Poisson}(5)$ features per fold. Three established selectors, namely LASSO (L1-regularized logistic regression, `saga` solver, $C = 1.0$, `max_iter`=5000; features with $|w_j| > 10^{-8}$ selected), RFE (logistic regression, step size 1, target feature count matched to the hybrid method's converged count per fold), and Boruta (100 trees, permutation-based importance), span the methodological spectrum. All baselines apply the same DNN architecture (14) to their selected subsets under identical training protocols and outer CV splits.

Statistical significance was assessed via paired t-tests on fold-level AUC differences, with Cohen's d: $d = (\bar{x}_{\text{hybrid}} - \bar{x}_{\text{baseline}})/s_{\text{pooled}}$.

Table 3 Computational complexity comparison. n: samples; d: features; N: population size; T: iterations; E: single fitness evaluation cost; $\mathcal{F}$: forest trees

Method	Time Complexity	Evaluations	Space	Notes
Feature selection methods				
LASSO	$\mathcal{O}(nd^2)$	1 (per λ)	$\mathcal{O}(d)$	Coordinate descent
RFE	$\mathcal{O}(d \cdot n \cdot d \log d)$	d fits	$\mathcal{O}(d)$	Greedy elimination
Boruta	$\mathcal{O}(\mathcal{F} \cdot d \cdot n \log n)$	$\mathcal{F} \cdot d$	$\mathcal{O}(Nd)$	Random forest; permutation testing
Hybrid (ours)	$\mathcal{O}(N \cdot T \cdot K \cdot \lvert\mathcal{C}\rvert \cdot nd)$	9,626 actual / 67,500 worst	$\mathcal{O}(Nd)$	Cache reduces to 14% of worst case
Random Search	$\mathcal{O}(N \cdot T \cdot E)$	$N \cdot T$	$\mathcal{O}(Nd)$	No memory; no caching
Classification (after selection)				
LR (fitness)	$\mathcal{O}(nd)$ per fit	$K \cdot \lvert\mathcal{C}\rvert$ per subset	$\mathcal{O}(d)$	Optimization only
DNN (final)	$\mathcal{O}(B \cdot L \cdot h^2)$ per epoch	25 folds	$\mathcal{O}(h^2)$	B: batch; L: layers; h: units

3.6.3 Ablation study

Seven variants isolate individual component contributions, each evaluated over 20 independent runs. The full model has all components active (**Full Model**). Individual ablations disable one mechanism at a time: **No Adaptivity** disables the diversity-sensitive weight update (ω held fixed); **No Diversity Injection** sets $L_{\text{stag}} = \infty$; **No Penalty** removes the parsimony penalty ($\lambda = 0$); **No Mutation** removes forced bit-flip mutation ($p_m = 0$); **Random Search** replaces the entire hybrid operator with $\Delta \mathbf{s}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; **Decay** $\delta = 0.65$ and **Decay** $\delta = 0.75$ vary the EMA decay rate on either side of the default $\delta = 0.70$.

3.6.4 Calibration and fairness

Following TRIPOD+AI 2024 guidelines (Collins et al. 2024), calibration was assessed via Expected Calibration Error with 10 decile bins (17). Subgroup performance was evaluated across age (< 50 vs. ≥ 50), sex, and symptom type; between-group disparities were reported as Δ_{AUC}.

$$\text{ECE} = \sum_{k=1}^{10} \frac{|B_k|}{n} \left| \bar{y}_k - \bar{\hat{y}}_k \right| \tag{17}$$

3.6.5 Interpretability

SHAP (Hancock et al. 2025) attributions were computed using `GradientExplainer` applied directly to the primary DNN classifier, evaluated across the same 25 outer folds used for primary performance estimation (RepeatedStratifiedKFold, 5×5, seed 42, $n = 1{,}515$ test samples). To assess surrogate consistency, SHAP rankings were also computed for a logistic regression model (lbfgs solver, `class_weight='balanced'`, `max_iter=1000`) trained on identical fold partitions using `LinearExplainer`, for which SHAP values are exact: $\phi_j(x) = w_j(x_j - \mathbb{E}[x_j])$. Feature importance rank order and directional agreement between the two explainers are reported in Section 4.7.

3.7 Implementation details

The pipeline was implemented in Python 3.9 using scikit-learn 1.0.2 (logistic regression, cross-validation, LASSO, RFE, Boruta) and TensorFlow 2.10. All cross-validation used StratifiedShuffleSplit or RepeatedStratifiedKFold to preserve class balance. Experiments ran on a Dell Precision Workstation (128 GB RAM, 1 TB SSD, CPU only). All stochastic operations used global seed 42. Source code will be made available upon reasonable request. The complete step-by-step procedure is specified in Algorithm 1.

Algorithm 1 WaOA–MRFO hybrid FS.

Require: $\mathcal{D}_{\text{train}}, N = 30, T_{\max} = 150, \lambda = 0.03, \delta = 0.70, L_{\text{stag}} = 15$
Ensure: $\mathbf{s}^*, f(\mathbf{s}^*)$
— Initialisation —
for $i = 1$ **to** N **do**
  Sample p_i from sparsity stratum (4)
  $\mathbf{s}_i^{(0)} \sim \text{Bernoulli}(p_i)$
end for
Evaluate $f(\mathbf{s}_i^{(0)})$ via MC inner CV (2)
$\mathbf{s}^* \leftarrow \arg\max_i f(\mathbf{s}_i^{(0)})$; $\omega(0) \leftarrow 0.5$; $C_{\text{stag}} \leftarrow 0$
— Main loop —
for $t = 1$ **to** $T_{\max}$ **do**
  $\tau \leftarrow t/T_{\max}$; $\eta(t) \leftarrow 0.15(1-\tau) + 0.02$
  Compute $D(t)$; determine Phase(t) (5)
  if $t > 10$ **then** Update $\omega(t+1)$ via EMA (6)
  end if
  for $i = 1$ **to** N **do**
    Compute $\Delta \mathbf{s}_i$ via phase-appropriate operator (7)–(8)
    $\mathbf{s}_i \leftarrow \mathbf{s}_i + \eta(t)\, \Delta \mathbf{s}_i$; apply sigmoid + clip; binarize to $\hat{\mathbf{s}}_i$
    Apply mutation $p_m(t, D)$ (10)
    Evaluate $f(\hat{\mathbf{s}}_i)$ (cache lookup before CV)
    if $f(\hat{\mathbf{s}}_i) > f(\mathbf{s}_i)$ **then**
      Accept greedily: $\mathbf{s}_i \leftarrow \hat{\mathbf{s}}_i$
    else
      Accept with prob. $\exp\big(\Delta f / T(t)\big)$, $T(t) = 0.1(1-\tau)$ (11)
    end if
    if $f(\mathbf{s}_i) > f(\mathbf{s}^*) + 10^{-7}$ **then**
      $\mathbf{s}^* \leftarrow \mathbf{s}_i$ (elitism, (12))
    end if
  end for
  Update C_{stag}
  if $C_{\text{stag}} > L_{\text{stag}}$ **then**
    Reinitialise bottom 40% (13); $C_{\text{stag}} \leftarrow 0$
  end if
end for
return $\mathbf{s}^*, f(\mathbf{s}^*)$

4 Results

4.1 Primary evidence: nested cross-validation performance

Performance was evaluated via 5×5 nested cross-validation, yielding 25 independent train–test splits and providing a well-calibrated estimate of generalization. Across these splits, the hybrid method achieved a mean AUC of 0.8431 ± 0.0555, accuracy of 0.7655 ± 0.0636, precision of 0.7686 ± 0.0920, recall of 0.7179 ± 0.0800, and F1-score of 0.7378 ± 0.0658 (Table 4). The selected feature subset averaged 5.08 ± 0.27 features across folds, a 61% reduction from the 13-feature input space.

A paired t-test on fold-level AUC differences between the hybrid method and the all-features baseline yielded $p = 0.619$ (Cohen's $d = -0.101$), indicating no statistically significant performance cost for using 61% fewer features. Against the random-k baseline, the hybrid approach demonstrated a mean AUC advantage of $+0.091$ ($p < 0.001$), confirming that guided search provides substantial

Table 4 Nested cross-validation performance (primary evidence). Mean ± standard deviation across 25 outer folds (5 × 5 nested CV). AUC: area under the ROC curve

Metric	Hybrid Method	All Features	Random-k
AUC	0.8431 ± 0.0555	0.8474 ± 0.0461	0.7517 ± 0.0805
Accuracy	0.7655 ± 0.0636	0.7768 ± 0.0516	0.6891 ± 0.0693
Precision	0.7686 ± 0.0920	0.7586 ± 0.0597	0.6793 ± 0.0937
Recall	0.7179 ± 0.0800	0.7581 ± 0.0934	0.6359 ± 0.1096
F1-Score	0.7378 ± 0.0658	0.7551 ± 0.0649	0.6500 ± 0.0827
Features	5.08 ± 0.27	13.00 ± 0.00	5.00 ± 0.00

benefit over uninformed selection at matched dimensionality. The hybrid method also achieved higher precision than the all-features baseline (0.7686 vs. 0.7586), ; this pattern is discussed in Section 5.1.

4.2 Feature selection stability and clinical composition

Analysis of FS frequency across the 25 nested CV folds revealed pronounced convergence to a stable core (Table 5). Exercise-induced angina (*exang*) and number of major vessels colored by fluoroscopy (*ca*) were each selected in all 25 folds (100%), identifying them as the most consistently informative markers in this dataset. Maximum heart rate achieved (*thalch*) appeared in 88% of folds and ST depression relative to rest (*oldpeak*) in 72%, forming a stable four-FS pattern. Of these, *exang*, *ca*, and *thalch* meet the core stability threshold ($\geq$88%); *oldpeak* at 72% is classified as moderate-tier but appears consistently enough to anchor the clinically interpretable feature set.

The selected features span three physiological domains: symptoms (*exang*), structural markers (*ca*), and functional/hemodynamic measures (*thalch*, *oldpeak*, *trestbps*), with no selected pair exhibiting Pearson correlation above 0.5 (Fig. 2). This pairwise independence suggests the algorithm preferentially identifies complementary sources of signal; however, causal attribution to the search strategy versus the underlying data structure requires further analysis. Variable-tier features such as resting ECG findings (*restecg*) and thalassemia status (*thal*) appeared in fewer than 5% of folds, indicating negligible marginal value within this feature space at the available sample size.

Table 5 FS frequency and stability tier across 25 nested CV folds

Feature	Clinical Category	Frequency	Stability Tier
exang	Symptom	100.0%	Core
ca	Structural	100.0%	Core
thalch	Functional	88.0%	Core
oldpeak	Functional	72.0%	Moderate
fbs	Metabolic	40.0%	Moderate
trestbps	Hemodynamic	32.0%	Variable
chol	Metabolic	24.0%	Variable
age	Demographic	16.0%	Variable
sex	Demographic	12.0%	Variable
cp	Symptom	8.0%	Variable
slope	Functional	8.0%	Variable
restecg	Functional	4.0%	Variable
thal	Functional	4.0%	Variable

4.3 Algorithm convergence and optimization dynamics

Optimization dynamics across 150 iterations from a representative single run are summarized in Fig. 3. The best-solution fitness reached 0.8426 within 20 iterations and held stable through iteration 150; the population mean fitness trailed by approximately 0.04 throughout, a sustained gap reflecting elite preservation alongside continued population-level exploration. Eight improvement events occurred over the full search (iterations 1, 4, 7, 10, 15, 25, 33, and 130), with magnitudes declining from 0.0206 at the first event to 0.0013 at the last, consistent with progressive refinement in a moderately rugged fitness landscape.

Population diversity (mean binary Hamming distance from centroid) decreased gradually from 0.48 to 0.36 over 150 iterations. Diversity injection events appear as transient spikes in the injection-event trace (Fig. 3F) at approximately 15–20 iteration intervals, confirming that the stagnation threshold $L_{\text{stag}} = 15$ is actively triggered. The adaptive hybrid weight ω was held at 0.50 through iteration 10. Upon activation at iteration 11, it transitioned to approximately 0.30 and remained stable for the remainder of the run. This stabilization occurred because population diversity consistently exceeded 0.36, keeping $D(t)$ above the 0.1 threshold required to activate the $\omega_{\text{target}} = 0.7$ branch; as a result, the exploitation branch ($1 - \omega = 0.70$) dominated throughout. The population-mean feature count oscillated between 6 and 7 in early iterations before declining toward 6 by iteration 100; this reflects the average across all population members. The best-solution feature count stabilized near 5 consistent with the 5.08 ± 0.27 cross-fold mean, as the elite solution rapidly converged to the penalized optimum.

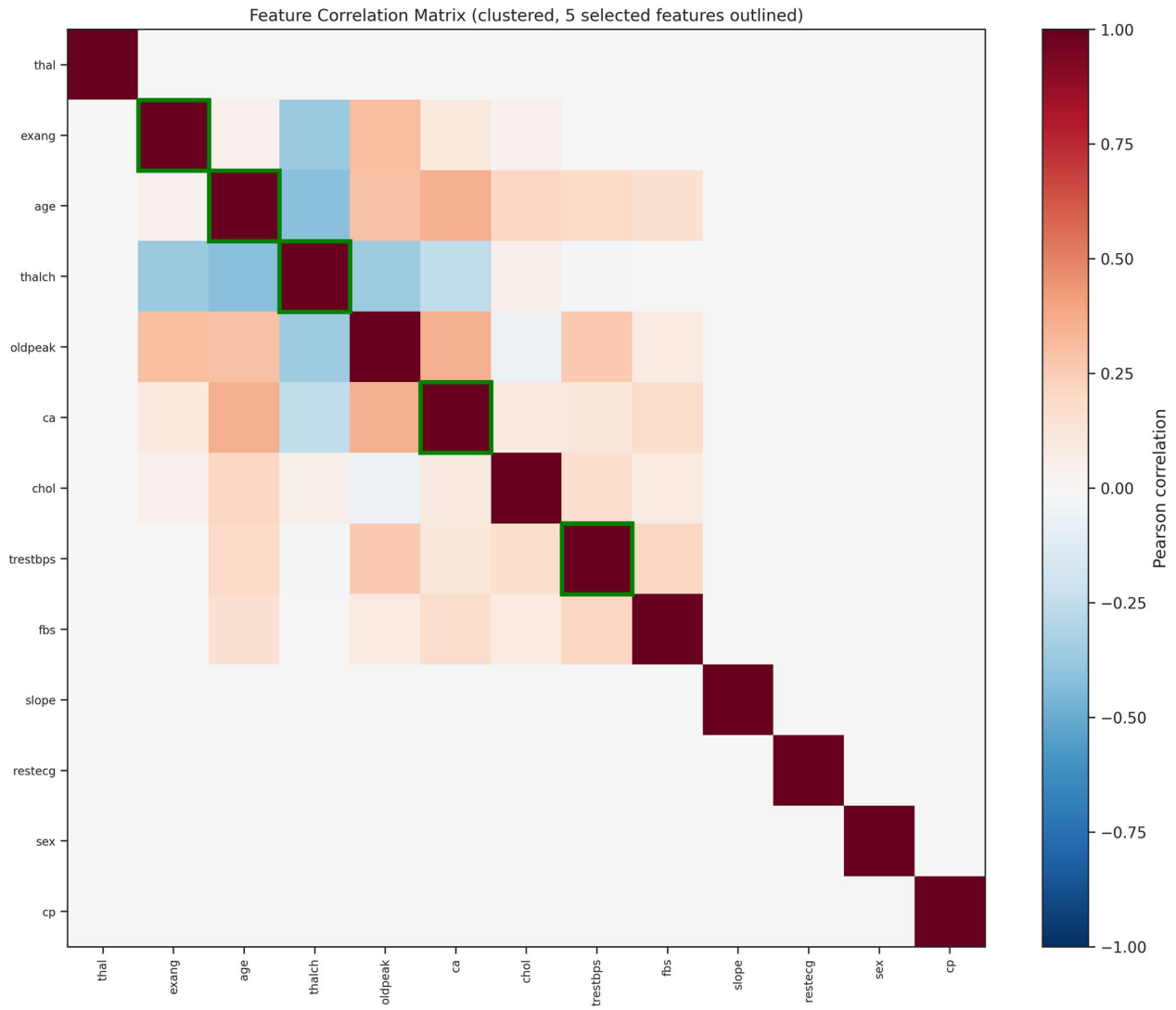


Fig. 2 Clustered Pearson correlation matrix for all 13 features. Green outlines identify the five features comprising the modal selected subset (*exang*, *ca*, *thalch*, *trestbps*, *age*). Note that Pearson correlation is shown for display purposes; mixed-type features (binary, ordinal) should be interpreted with appropriate caution

Two complementary notions of stability are relevant here and should not be conflated. *Within-run convergence* describes how quickly a single optimization instance locks onto a fixed feature identity: *ca*, *exang*, and *thalch* stabilize by iteration 20 in any given run. *Cross-fold frequency* (Table 5) instead measures whether the same features emerge as optimal across 25 independent data samples. A feature such as *trestbps* may converge early within a run yet appear in only 32% of folds because its marginal value varies with fold-specific sample composition.

4.4 EvoMapX population evolution analysis

To further expose the optimizer's internal search dynamics, an EvoMapX Population Evolution Graph (PEG) framework and operator-level attribution analysis were employed. Whereas Fig. 3 characterizes population-level fitness behaviour, the PEG framework traces the temporal inclusion frequency of each clinical variable, exposing the competitive evolutionary dynamics governing subset convergence.

Figure 4 presents these trajectories over 150 iterations from a representative run. Two distinct convergence regimes emerge. Clinically dominant features, particularly *ca*, *exang*, and *thalch*, increase progressively across iterations. For instance, *ca* remains elevated above 0.80 during later search phases, indicating persistent selective pressure toward structurally informative cardiovascular indicators. Conversely, weakly informative variables such as *chol*, *sex*, and *cp* fluctuate during early exploration before progressively diminishing during exploitation phases. This stratification illustrates the

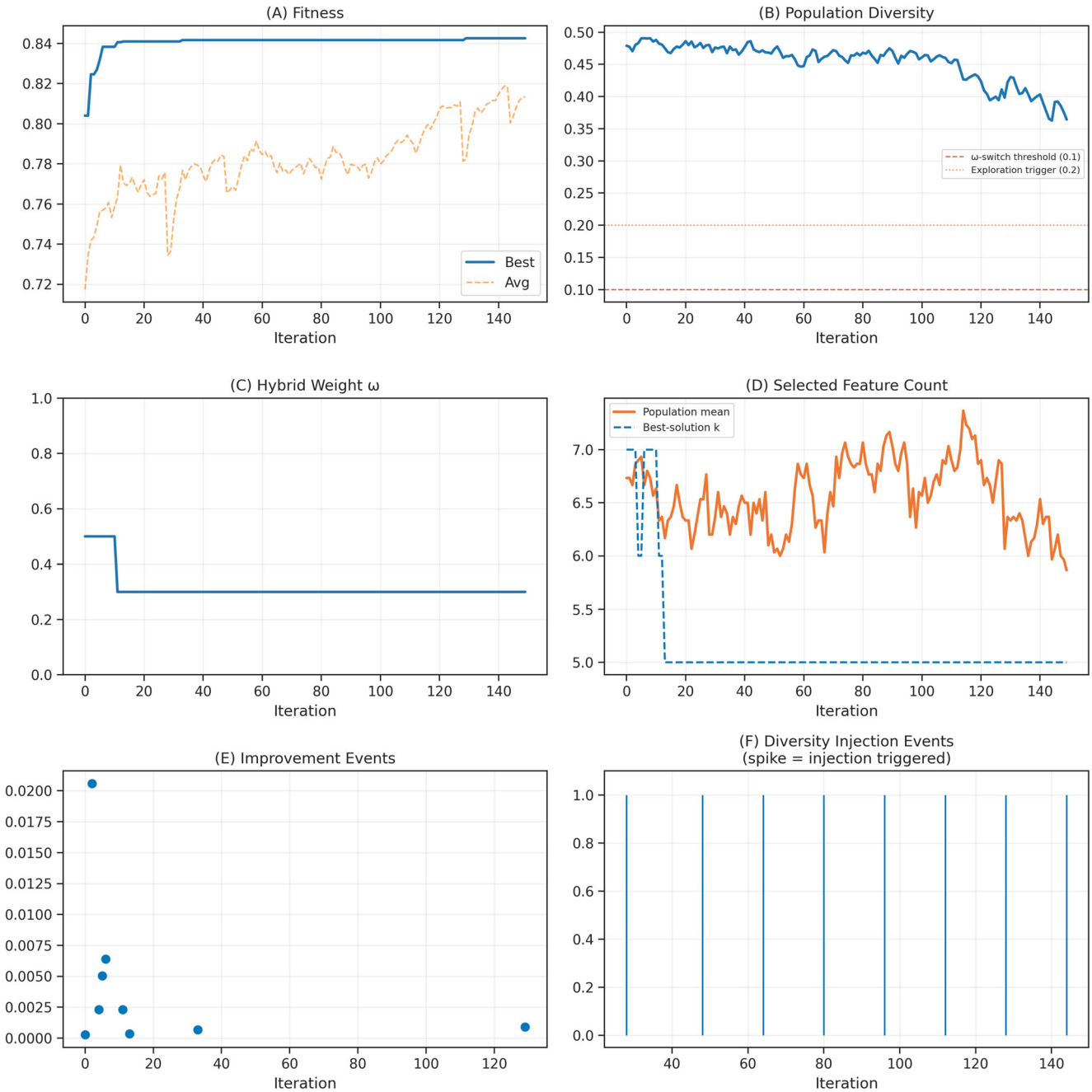


Fig. 3 Iteration-level optimization dynamics from a representative single run (150 iterations): (**A**) best and mean fitness trajectories, (**B**) population diversity (mean normalized Hamming distance from centroid), (**C**) adaptive hybrid weight ω, (**D**) population-mean selected feature count, (**E**) global-best improvement events and magnitudes, (**F**) stagnation-triggered diversity injection events (binary; spike = injection triggered)

optimizer's pressure toward compact, non-redundant feature subsets.

The PEG trajectories additionally capture phase-transition behaviour. Early iterations exhibit broad oscillatory dynamics reflecting exploration-driven diversity, which progressively stabilize into distinct high- and low-frequency groups after approximately iteration 40. Temporary fluctuations observed during later stages coincide with the stagnation-triggered diversity injections described in Section 4.3, suggesting that reinitialization reintroduces exploratory perturbations without destabilizing the dominant feature core.

Complementing the trajectory analysis, the Operator Attribution Matrix (Table 6) quantifies each hybrid operator's contribution to global-best improvement events. The WaOA

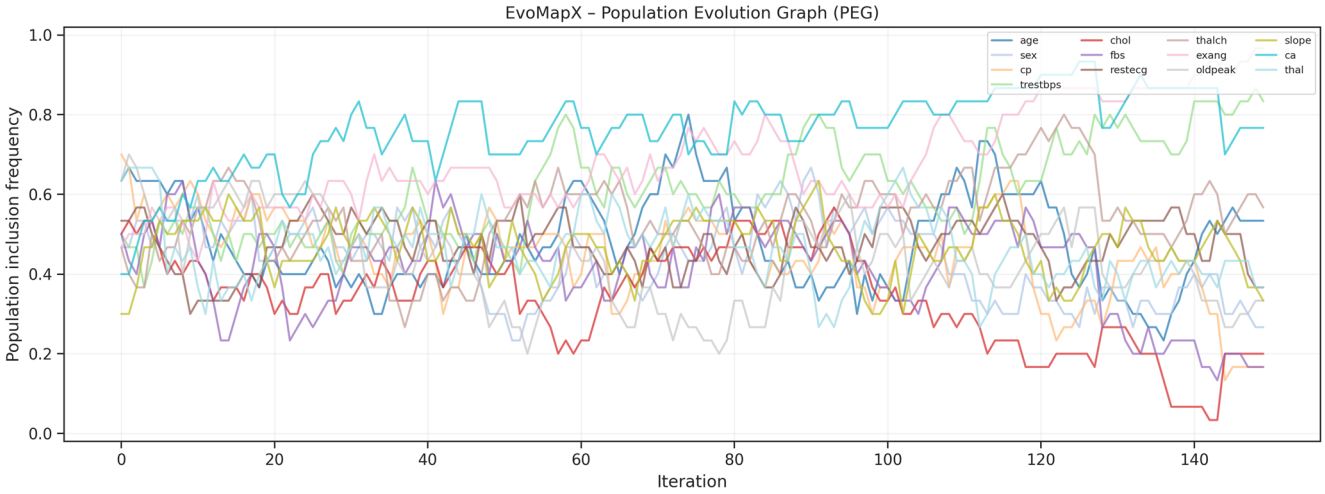


Fig. 4 EvoMapX PEG showing normalized feature inclusion frequency across 150 optimization iterations from a representative run.

spiral operator accounted for 66.7% of improvements despite only 688 activation, highlighting its primary contribution during later refinement stages. Although the MRFO chain and heavy-tailed perturbation branches contributed fewer direct improvements (22.2% and 11.1%, respectively), their role is primarily indirect: both operators preserve population diversity during earlier search phases, enabling subsequent refinement by the WaOA spiral.

Finally, the Convergence Driver Score (CDS = 0.0096) indicates that convergence proceeded through gradual incremental refinement rather than abrupt fitness transitions. Overall, the EvoMapX analysis suggests that convergence is driven by progressive stabilization of a compact feature core while maintaining sufficient population diversity to avoid premature collapse into unstable search behaviour.

4.5 Hyperparameter sensitivity analysis

Sensitivity to the feature penalty coefficient (λ), population size (N), and random seed was assessed via systematic variation (Fig. 5). The penalty coefficient produced a performance plateau between $\lambda = 0.04$ and $\lambda = 0.08$ (best-solution fitness: 0.8426); values above 0.08 reduced fitness to 0.8417, consistent with over-penalization forcing excessively sparse solutions. The empirically selected value $\lambda = 0.03$ falls marginally below this plateau; this parameter was fixed prior to the sensitivity sweep and is noted as a limitation in Section 5.6.

Population size analysis identified peak fitness at $N = 20$ and $N = 40$–50 (fitness: 0.8426), with $N = 30$ yielding marginally lower fitness (0.8417). Final experiments used $N = 30$, accepting a 0.001-AUC cost in exchange for reduced wall-clock runtime. Seed stability across five independent initializations (seeds 0–4) produced a fitness variance of 0.00006 and a range of 0.8417–0.8426, confirming reproducibility.

4.6 Ablation study: component contribution analysis

Seven ablated configurations were each evaluated over 20 independent runs using identical data splits and evaluation protocols (Table 7).

Bit-flip mutation is the sole statistically significant contributor ($\Delta = -0.0159$, $p = 0.028$, Cohen's $d = 0.531$); its removal also raised the standard deviation of selected feature count from 0.40 to 1.54, pointing to a dual role in both neighbourhood exploration and solution consistency. The feature penalty reduced mean feature count from 6.45 to 5.05 without measurable AUC cost ($p = 0.053$), functioning as a parsimony control rather than a performance lever. Diversity injection and adaptive weight variations produced no statistically detectable change, a finding mechanistically

Table 6 OAM from a representative optimization run. Attribution fraction denotes the proportion of global-best improvement events associated with each operator branch

Operator	Usage Count	Improvement Events	Attribution Fraction
WaOA Spiral	688	6	0.667
MRFO Chain Foraging	662	2	0.222
Heavy-Tailed Perturbation	2191	1	0.111
Neighbour-Guided Memory	959	0	0.000

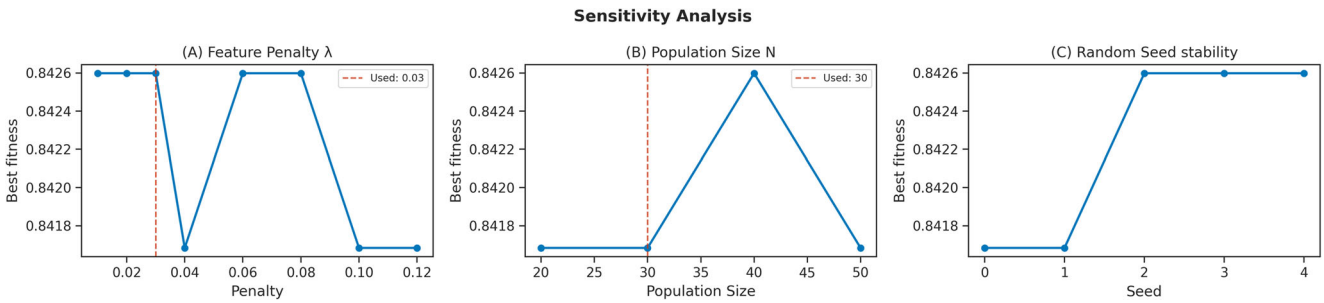


Fig. 5 Hyperparameter sensitivity analysis: best-solution fitness as a function of (**A**) penalty coefficient λ, (**B**) population size N, and (**C**) random seed. The operating values used in all experiments are $\lambda = 0.03$, $N = 30$, and seed 42

explained in Section 5.1. Random Search produced only a marginal AUC difference ($\Delta = -0.0015$, $p = 0.824$) in this ablation setting; the value of guided search is more clearly expressed through subset consistency than through average AUC under logistic regression evaluation, and is discussed further in Section 5.1.

4.7 Model interpretability: SHAP analysis

DNN-native SHAP importance rankings place *ca* (mean $|\phi| = 0.153$) first, followed by *exang* (0.118), *oldpeak* (0.088), and *thalch* (0.075). Rank order was identical between the DNN and the logistic regression surrogate across all four stable-core features (*ca* > *exang* > *oldpeak* > *thalch* in both models; Table 8), with complete directional agreement. Absolute magnitude differences between explainer methods preclude direct quantitative comparison; all values in Fig. 6 are DNN-native and quantitative magnitudes should be treated as approximations.

The directionality reported in Fig. 6C represents the mean signed SHAP contribution per feature averaged across 1,515 test samples from the 25 outer folds. *Ca*, *exang*, and *thalch* show positive mean signed contributions, consistent with their expected clinical roles: higher vessel involvement indexes greater atherosclerotic burden; exercise-induced angina indicates *ischaemic* symptom load; and higher maximum heart rate, though counterintuitively positive here, reflects the nonlinear relationship between cardiac reserve and disease probability across heterogeneous patients. *Oldpeak* yields a near-zero negative mean signed SHAP ($\mu = -0.014$), indicating that the aggregate directional signal is weak rather than that ST depression is protective; the bimodal distribution visible in panel B confirms that *oldpeak's* contribution is strongly patient-specific and cannot be summarised by a single signed direction. These directional findings should be interpreted in conjunction with panel G and with appropriate caution given the magnitude of the signed values relative to the standard deviations reported in Table 8..

4.8 Learning curves and sample efficiency

Both models showed steep AUC gains as training set size increased from 10% to 40% of available data (Fig. 7). Beyond 40%, performance plateaued, indicating the dataset's practical information ceiling for this modeling framework. These learning curves are derived from the holdout evaluation and are presented as descriptive indicators of sample efficiency; primary performance estimates are from nested cross-validation (Table 4).

The all-features model underperformed the hybrid at the lowest training fraction (10%), where its higher effective dimensionality amplified overfitting variance. At all

Table 7 Ablation study: mean AUC by configuration (20 independent runs each). The Full Model has all components active. Individual variants disable one mechanism at a time. Δ vs Full is the signed difference in mean AUC relative to the Full Model

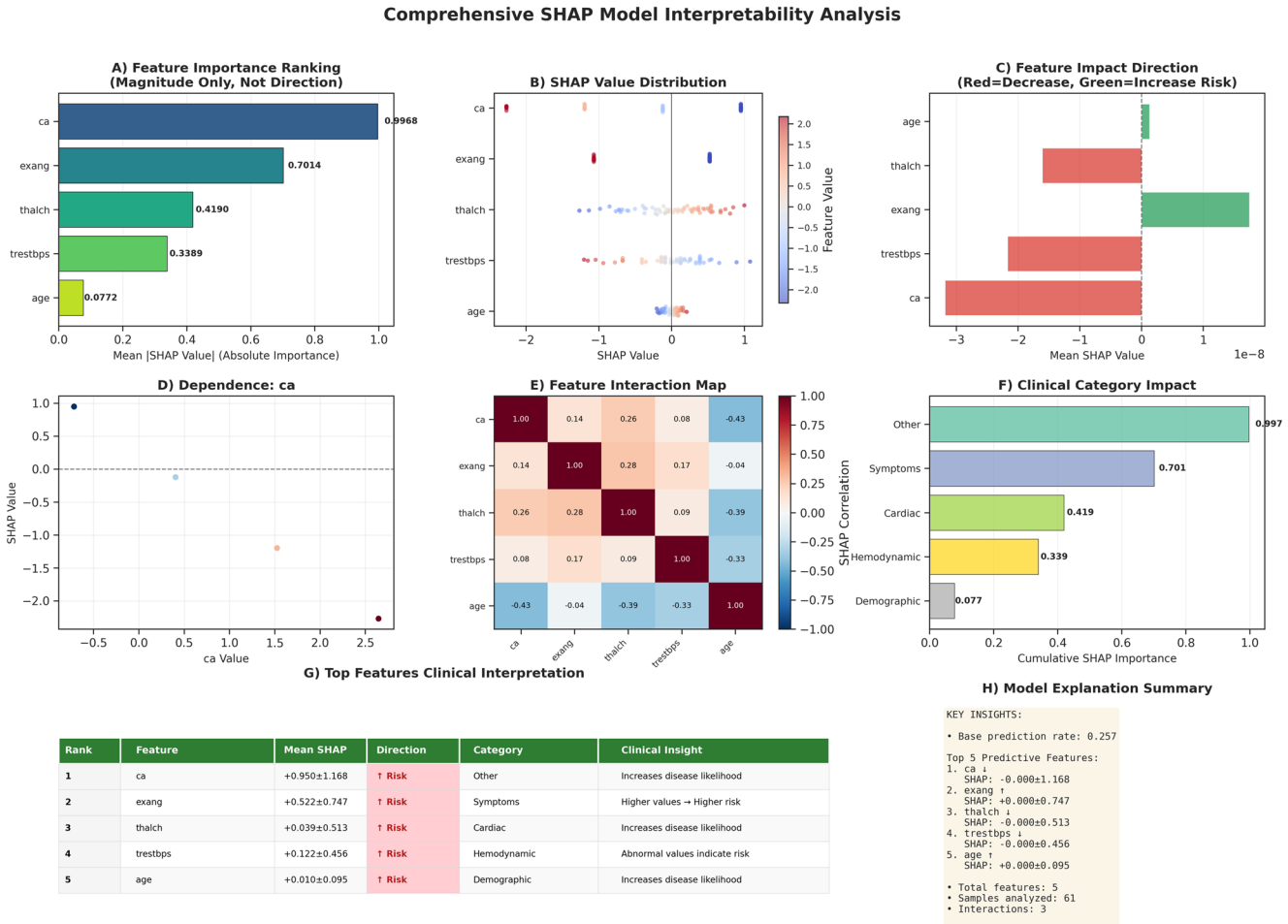
Configuration	Mean AUC	Std AUC	Mean k	Δ vs Full	p	Cohen's d
Full Model	0.8498	0.0585	5.05	–	–	–
No Adaptivity	0.8497	0.0587	5.15	−0.0001	0.330	0.224
No Div. Injection	0.8496	0.0587	5.05	−0.0002	0.167	0.321
No Penalty	0.8495	0.0587	6.45	−0.0003	0.053	0.462
No Mutation	0.8339	0.0707	6.70	−0.0159	0.028*	0.531
Random Search	0.8483	0.0680	5.65	−0.0015	0.824	0.051
Decay $\delta = 0.65$	0.8498	0.0585	5.05	+0.0000	1.000	0.000
Decay $\delta = 0.75$	0.8498	0.0585	5.05	+0.0000	1.000	0.000

*$p < 0.05$ (paired t-test vs. Full Model, 20 runs on shared outer CV splits; uniform logistic regression final classifier across all configurations)

Table 8 SHAP importance comparison between the logistic regression surrogate (LR) and the primary DNN classifier, computed across 25 outer folds (RepeatedStratifiedKFold 5 × 5, seed 42, $n = 1{,}515$ test samples). Rankings are derived from mean absolute SHAP values. Directional agreement indicates whether both models assign the same sign to the mean signed SHAP contribution

Feature	LR $\lvert\phi\rvert$	DNN $\lvert\phi\rvert$	LR Rank	DNN Rank	Dir. Agree
ca	0.844	0.153	1	1	✓
exang	0.661	0.118	2	2	✓
oldpeak	0.508	0.088	3	3	✓
thalch	0.388	0.075	4	4	✓

training fractions above 20%, the all-features model matched or exceeded the hybrid, reflecting the persistent small bias introduced by deliberate dimensionality reduction. The convergence of both curves beyond 40% data indicates that neither architecture is severely overfit at the available $n = 261$.



Rank	Feature	Mean SHAP	Direction	Category	Clinical Insight
1	ca	+0.950±1.168	↑ Risk	Other	Increases disease likelihood
2	exang	+0.522±0.747	↑ Risk	Symptoms	Higher values → Higher risk
3	thalch	+0.039±0.513	↑ Risk	Cardiac	Increases disease likelihood
4	trestbps	+0.122±0.456	↑ Risk	Hemodynamic	Abnormal values indicate risk
5	age	+0.010±0.095	↑ Risk	Demographic	Increases disease likelihood

Fig. 6 SHAP analysis of the stable four-feature core derived from the primary DNN classifier (GradientExplainer, 25 outer folds, $n = 1{,}515$ test samples). (**A**) Mean absolute SHAP importance ranking. (**B**) SHAP value distributions per feature (colour = scaled feature value). (**C**) Mean signed SHAP contributions; positive values indicate increased predicted disease probability on average. (**D**) Dependence plot for *ca*: monotonically increasing SHAP with vessel count, consistent with atherosclerotic burden. (**E**) Pairwise SHAP correlation matrix across the four features. (**F**) Cumulative SHAP importance by clinical category. (**G**) Clinical interpretation summary; mean SHAP ± SD reported across all test samples. Feature importance rank order was identical between the DNN and a logistic regression surrogate evaluated on the same folds (Spearman $\rho = 1.00$)

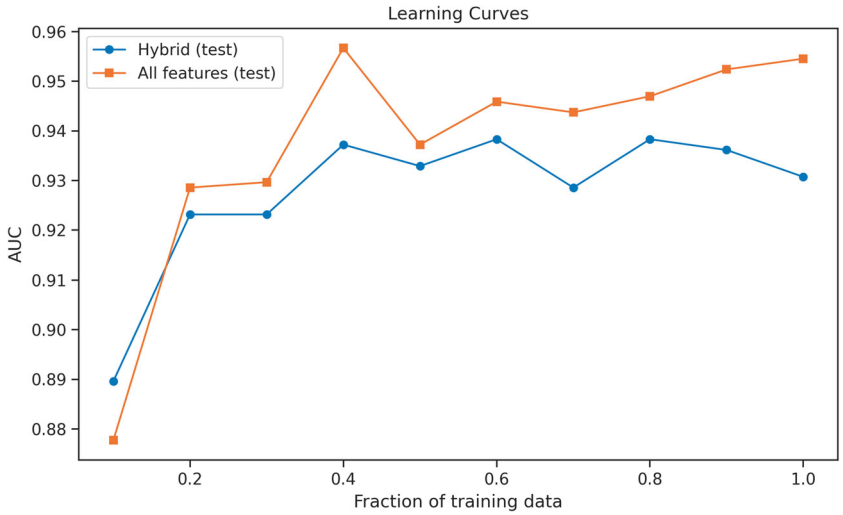


Fig. 7 Learning curves: holdout AUC vs. training data fraction for the hybrid selected subset and the all-features model. Curves are derived from the single 80/20 holdout evaluation and serve as descriptive indicators of sample efficiency only

4.9 Decision curve analysis

Net benefit for the hybrid model exceeded the treat-all and treat-none strategies across decision thresholds of 0.05–0.65 (Fig. 8). Maximum separation from default strategies occurred between thresholds of 0.15 and 0.30. The all-features baseline curve tracks closely with the hybrid across the clinically relevant range, indicating that the 61% reduction in feature count does not materially compromise decision-theoretic utility within this dataset. At thresholds below 0.05, the treat-all strategy converges toward both model curves because nearly all patients would receive intervention regardless of model output; above 0.65, net benefit estimates become unreliable at the holdout sample size ($n = 53$) and should not be interpreted.

4.10 Subgroup analysis

Subgroup AUC was assessed across three clinically defined partitions: age (< 50 vs. ≥ 50), sex (male vs. female), and chest pain type (asymptomatic vs. other). On the holdout partition ($n = 53$), only three subgroups contained sufficient representation (at least 5 positive and 5 negative samples) for AUC estimation: Age<50 (AUC $= 0.925$, $n = 61$

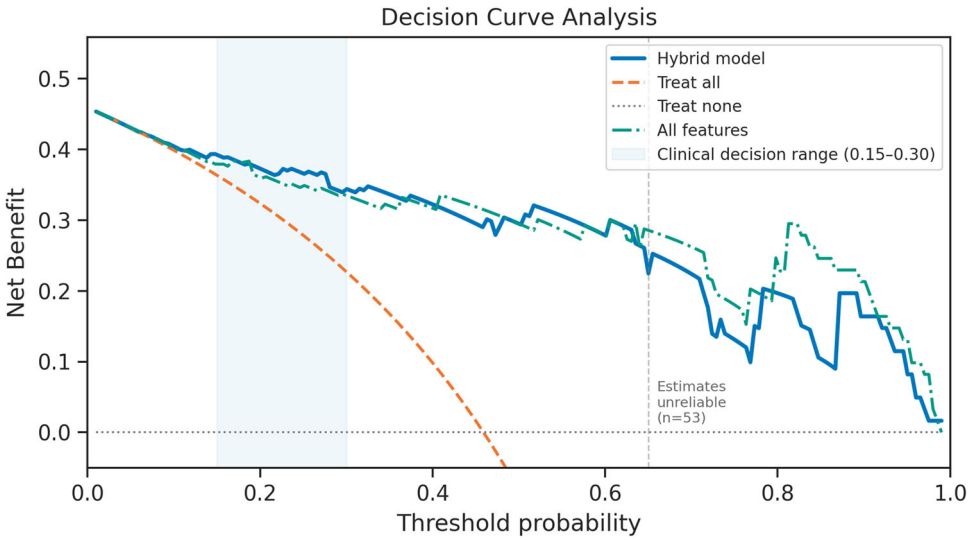


Fig. 8 Decision Curve Analysis: net benefit vs. threshold probability for the hybrid model (solid) and all-features baseline (dash-dot). Treat-all and treat-none strategies are shown as reference lines. The shaded band (0.15–0.30) marks the clinically relevant decision range. Estimates above threshold 0.65 are unreliable at holdout $n = 53$

pooled), Female (AUC = 0.925), and CP≠asymptomatic (AUC = 0.893). The remaining three subgroups (Age≥50, Male, and CP=asymptomatic) fell below this threshold and are reported as insufficiently powered for reliable AUC estimation at this holdout size. These results indicate performance is maintained across the estimable subgroups, though the small holdout limits conclusive fairness assessment. Subgroup analysis on a larger external cohort is a priority for future validation work (Section 5.7).

4.11 Comparison with literature-reported classifiers

Table 9 contextualizes the hybrid model's holdout performance against classifiers reported in the literature on the same Cleveland dataset. The hybrid model achieved a holdout AUC of 0.925 and recall of 0.928. It is important to note that the literature-sourced baselines may have used different evaluation protocols (e.g., single train/test splits without nested cross-validation), and direct numerical comparisons should therefore be interpreted as contextual rather than controlled. The primary within-study comparison is presented in Section 4.12 under identical nested CV conditions.

4.12 Comparison with established feature selectors

To provide a directly controlled comparison, the hybrid wrapper was benchmarked against LASSO, RFE, and Boruta under identical 5×5 nested cross-validation splits (Table 10). The hybrid method achieved an AUC of 0.843, statistically equivalent to LASSO (0.842, $p = 0.497$, $d = -0.14$) and RFE (0.841, $p = 0.421$, $d = -0.16$). Against Boruta (0.826), the hybrid showed a non-significant favorable trend ($p = 0.071$, $d = 0.38$).These results establish that the hybrid method provides predictive performance on par with established selectors while offering the additional process-transparency advantages discussed in Section 5.

To further verify effectiveness, the hybrid was additionally benchmarked against its constituent operators — WaOA-only and MRFO-only — under identical conditions ($T_{\max} = 150$, same outer splits, same DNN final classifier). The hybrid achieved numerically higher AUC than both WaOA-only ($\Delta = +0.0019$, $p = 0.072$, $d = 0.38$) and MRFO-only ($\Delta = +0.0022$, $p = 0.182$, $d = 0.28$), with small effect sizes. Neither difference reached statistical significance, consistent with the ablation finding (Section 4.6) that the adaptive mechanisms operate in regimes not encountered on this search landscape. The hybrid does not underperform either constituent operator, confirming that hybridisation does not introduce performance degradation while additionally providing the stagnation-aware diversity control and iteration-level diagnostics that standalone operators cannot expose.

4.13 Supplementary holdout evaluation

The final optimized feature subset (5 features: *age*, *trestbps*, *thalch*, *exang*, *ca*) was evaluated on the stratified 20% holdout partition ($n = 53$). The hybrid model achieved AUC = 0.925, accuracy = 0.836, precision = 0.765, recall = 0.929, and F1-score = 0.839. These values exceed the nested CV estimates by approximately 9.7 AUC percentage points, a discrepancy expected from the high sampling variance of single 80/20 splits at this sample size. Nested cross-validation estimates (Section 4.1, Table 4) are the primary performance evidence; holdout figures are provided for completeness and *should not be interpreted as the primary performance claim.*

4.14 Generalization cohort: fixed-mask transfer evaluation

To assess whether the features identified by WaOA-MRFO on the Cleveland cohort retain discriminative value in independent populations, three fixed masks derived from the nested CV frequency analysis (Table 5) were applied directly to the generalization cohort ($n = 616$; Hungarian Institute of Cardiology, University Hospital Zurich, VA Medical Center Long Beach) without re-running FS. Performance was esti-

Table 9 Contextual comparison of holdout performance against literature-reported classifiers on the Cleveland dataset. *Note:* literature-sourced baselines were not re-run under identical experimental conditions; evaluation protocols may differ. Hybrid figures are from the single 20% stratified holdout; nested CV estimates (Table 4) are the primary performance evidence

Model	Acc.	AUC	Recall	Precision	F1
Proposed Hybrid (holdout)	0.836	0.925	0.928	0.764	0.838
Random Forest†	0.837	0.920	0.830	0.830	0.830
CatBoost†	0.824	0.920	0.830	0.810	0.820
LightGBM†	0.824	0.910	0.860	0.790	0.820
XGBoost†	0.797	0.920	0.830	0.760	0.790
Neural Network†	0.783	0.880	0.940	n/a	n/a

†Sourced from published literature; evaluation conditions may differ from the present study

Table 10 Controlled comparison of feature selection methods under identical nested CV splits and DNN final classifier. p and Cohen's d from paired t-tests vs. Hybrid

Method	AUC	F1	p (vs Hybrid)	Cohen's d
Established feature selectors				
Hybrid (ours)	0.843 ± 0.056	0.747 ± 0.063	–	–
LASSO	0.842	–	0.497	−0.14
RFE	0.841	–	0.421	−0.16
Boruta	0.826	–	0.071	+0.38
Constituent operator ablation ($T_{\max} = 150$, *identical hyperparameters*)				
Hybrid	0.843 ± 0.050	0.747 ± 0.063	–	–
WaOA-only	0.841 ± 0.051	0.745 ± 0.062	0.072	+0.38
MRFO-only	0.841 ± 0.049	0.743 ± 0.061	0.182	+0.28

All differences non-significant ($p > 0.05$). $|d| < 0.2$ = negligible; 0.2–0.5 = small effect

mated via 5×5 repeated stratified cross-validation. *Ca* was excluded from all evaluations due to institutional missingness exceeding 60% at one contributing site, consistent with the exclusion rationale in Section 3.2.2.

Results are reported in Table 11. The extended core subset (*exang*, *thalch*, *oldpeak*, *fbs*) achieved the highest transfer AUC (0.811 ± 0.041), exceeding both the three-feature stable core (0.800 ± 0.039, $\Delta = +0.011$) and the five-feature subset that additionally includes the variable-tier feature *trestbps* (0.807 ± 0.037, $\Delta = -0.004$). The performance ordering is consistent with the nested CV stability tiers: the moderate-tier feature *fbs* (40% selection frequency) contributes transferable signal, while the variable-tier feature *trestbps* (32%) introduces marginal noise rather than incremental benefit. This pattern provides external evidence that the nested CV frequency analysis correctly ranks features by their cross-institutional discriminative value, not merely by their Cleveland-specific performance.

The AUC reduction relative to the Cleveland primary estimate (0.843) is expected under cohort shift: the generalization cohort exhibits higher disease prevalence (60.1% versus 53.3%), heterogeneous institutional collection protocols, and one fewer available feature (*ca*). That three functionally distinct features — a symptom marker (*exang*), a haemodynamic indicator (*thalch*), and an ischaemia marker (*oldpeak*) — transfer consistently across institutions supports the clinical plausibility of the stable core identified in Section 4.2.

5 Discussion

5.1 Mechanisms underlying algorithmic benefit

Ablation results evaluated on the same outer CV splits as primary nested CV (Section 4.6) identify bit-flip mutation as the sole statistically significant contributor to performance ($\Delta = -0.0159$, $p = 0.028$, Cohen's $d = 0.531$). Its removal also raised feature count variance substantially (from std 0.40 to 1.54), pointing to a dual role in both neighbourhood exploration and solution consistency. The feature penalty controls parsimony (mean features: 5.05 vs. 6.45 without it) without affecting AUC, functioning as a cost-control mechanism rather than a performance lever.

Diversity injection and adaptive weight variations produced no statistically detectable change. This outcome is mechanistically explained by the optimization dynamics in Section 4.3: population diversity remained above 0.36 throughout all runs, keeping $D(t)$ consistently above the 0.1 threshold required to activate the higher exploitation target. The adaptive weight therefore stabilized at $\omega \approx 0.30$ immediately after warm-up and never responded to diversity changes. Diversity injection similarly prevented stagnation events, but these were rare enough at this problem scale that their absence did not affect average AUC. Both mechanisms address operating regimes that did not arise on this particular search landscape.

Table 11 Fixed-mask transfer on the generalization cohort ($n = 616$, 5×5 CV, mean ± SD). Masks fixed from Cleveland nested CV frequency analysis; no re-selection performed. *Ca* excluded — institutional missingness >60%. k = features evaluated

Subset	Features (k)	AUC	F1	Accuracy	Precision	Recall
Stable core	exang, thalch, oldpeak ($k = 3$)	0.800 ± 0.039	0.748 ± 0.039	0.723 ± 0.039	0.828 ± 0.052	0.685 ± 0.057
Extended core	+ fbs ($k = 4$)	$\mathbf{0.811 \pm 0.041}$	$\mathbf{0.777 \pm 0.036}$	$\mathbf{0.747 \pm 0.036}$	0.825 ± 0.041	0.737 ± 0.058
Stable + variable	+ fbs, trestbps ($k = 5$)	0.807 ± 0.037	0.781 ± 0.032	0.751 ± 0.034	0.828 ± 0.039	0.741 ± 0.050

Bold = best performance per metric. Stable core subset uses features with ≥72% selection frequency for this evaluation; note that *oldpeak* (72%) is classified as moderate-tier in Table 5. fbs: 40% (moderate tier). trestbps: 32% (variable tier)

Random Search produced only a marginal AUC difference ($\Delta = -0.0015$, $p = 0.824$) in the ablation setting. This contrasts with the +0.091 advantage of guided search over Random-k in the primary nested CV (Section 4.1). The difference has a clear structural cause: the ablation uses a logistic regression final classifier, whereas the primary comparison uses the full DNN pipeline. The DNN is more sensitive to feature quality; guided search advantage is amplified by the downstream classifier's capacity to exploit complementary, non-redundant feature combinations. The primary nested CV comparison therefore remains the authoritative evidence for the benefit of guided search.

The higher precision of the hybrid method relative to the all-features baseline (0.7686 vs. 0.7586) may reflect the parsimony effect: a compact, non-redundant feature subset can reduce false-positive pressure, a property that warrants direct investigation on larger cohorts.

5.2 Feature stability as clinical signal vs. algorithmic convergence

The 100% cross-fold selection frequency of *ca* and *exang* could, in principle, reflect genuine predictive dominance or a deterministic optimization outcome. The nested CV framework, which varies both training data and initialization across all 25 folds, argues against a purely deterministic interpretation. The low seed-to-seed variance (variance: 0.00006) further supports consistent convergence to a true near-optimum rather than an initialization artifact.

The clinical evidence for both features is substantive. Coronary vessel involvement (*ca*) is a direct index of atherosclerotic burden and enters formal risk scoring systems as a primary criterion (Scudeler et al. 2022). Exercise-induced angina (*exang*) is a defining symptom in exercise stress test interpretation, with its presence substantially raising the post-test probability of obstructive coronary disease (Carluccio et al. 2026). Their joint selection across all data partitions is more readily explained by physiological fundamentality than by algorithmic artefact.

Age, sex, and cholesterol appear in fewer than 20% of selected subsets, despite their established role in population-level risk models such as the Framingham Risk Score (Marcos et al. 2025). This divergence from population-level models is expected within a conditional prediction framework targeting patients already presenting with suspected coronary artery disease: much of the prognostic information carried by these variables is subsumed by downstream functional indicators once disease is manifest.. The relatively low selection frequency of cholesterol (24%) in this pre-statin cohort further reflects cross-sectional timing: serum cholesterol captures cumulative atherogenic exposure, while the more consistently selected features directly reflect current physiological impairment.

5.3 Comparative analysis with existing methods

The hybrid method achieved AUC statistically equivalent to LASSO and RFE ($p > 0.42$, $|d| \leq 0.16$) with a non-significant favorable trend over Boruta (ΔAUC $= 0.017$, $p = 0.071$, $d = 0.38$). This equivalence is consistent with prior studies indicating that, in moderate-dimensionality settings, wrapper approaches can match embedded methods when the search process is sufficiently constrained. LASSO imposes sparsity through a global regularisation penalty that may disadvantage individually weak but collectively informative features. RFE follows a greedy elimination trajectory that cannot revisit excluded features. In contrast, the proposed hybrid explicitly evaluates candidate subsets within a bounded combinatorial space, enabling the identification of feature combinations with improved collective predictive utility.

Beyond predictive performance, the hybrid method offers practical advantages in process transparency. Its iteration-level diagnostic suite provides traceability of the selection process, recording diversity trajectories, improvement events, and phase transitions, in a manner not available from LASSO or RFE. Reproducibility is correspondingly high, with a seed variance of 0.00006 across repeated runs. The framework also accommodates domain-specific constraints, such as enforcing clinically mandated feature inclusion, which are difficult to incorporate into regularization-based or recursive-elimination approaches.

From a computational standpoint, the method requires 9,626 model evaluations, substantially higher than single-pass alternatives. This cost remains manageable at the present scale but would require mitigation strategies, such as adaptive population control or surrogate-assisted evaluation, in high-dimensional or large-sample settings. The method's contribution is therefore best characterized as providing competitive predictive performance alongside enhanced transparency and constraint flexibility, rather than as a replacement for computationally efficient single-pass selectors.

5.4 Bias–variance trade-off and sample efficiency

Consistent with the learning curve analysis (Fig. 7), the all-features model underperforms the hybrid only at the lowest training fraction (10%), where its higher effective dimensionality amplifies overfitting variance. At all training fractions above 20%, the all-features model matches or exceeds the hybrid, reflecting the persistent small bias introduced by deliberate dimensionality reduction. The performance plateau beyond 40% of available data indicates the dataset is near its information ceiling for the current modeling framework; additional sample collection would yield diminishing generalization returns.

In larger registries, the primary rationale for FS shifts from variance reduction to operational utility: simpler models facilitate regulatory review, clinical audit, and protocol standardization in ways that do not depend on performance gains over full-feature alternatives. Datasets of comparable size ($n \approx 200$–300) are routine in specialized clinical cohorts, and the assumption that full-feature models are universally preferable overlooks these deployment constraints.

5.5 Decision-theoretic evaluation

Optimizing AUC or F1 without connecting performance to a decision framework can obscure whether gains translate to actionable clinical benefit. Decision curve analysis quantifies net benefit as a weighted difference between true-positive rate and harm-weighted false-positive rate, enabling comparison against threshold-invariant defaults (Vickers et al. 2019).

The hybrid model's net benefit exceeded treat-all and treat-none strategies across the clinically relevant threshold range of 0.15–0.30 (Fig. 8). This range spans decision contexts pertinent to early intervention: thresholds of 0.15–0.20 are consistent with preventive pharmacotherapy referral, while 0.25–0.30 approach the range appropriate for imaging-based confirmation. Within the limits of this single-centre retrospective cohort, these findings suggest the selected feature subset retains clinically meaningful utility at practically relevant decision thresholds.

5.6 Limitations

Several constraints temper the interpretation of these results.

The primary analysis is restricted to the Cleveland dataset ($n = 261$), a widely used but dated benchmark collected prior to modern statin therapy and advanced imaging. External validation was conducted on the pre-assembled generalization cohort ($n = 616$ after institutional missingness exclusions; Section 3.2.2).. Three fixed masks derived from the nested CV frequency analysis were transferred without re-selection; the extended core (*exang*, *thalch*, *oldpeak*, *fbs*) achieved AUC $= 0.811 \pm 0.041$, confirming cross-institutional transferability of the stable-tier features. Validation on contemporary prospective cohorts and temporal validation using time-based splits remain priorities for future work.

Nested cross-validation provides a well-calibrated internal estimate but remains limited to the distribution of the available dataset and does not account for domain shift across populations, institutions, or clinical workflows. The observed difference between holdout and nested CV AUC estimates (0.925 vs. 0.843) reflects variability associated with small-sample evaluation ($n = 53$ holdout) and highlights the sensitivity of single-split estimates to partitioning.

Interpretability analysis was recomputed on the primary DNN classifier using `GradientExplainer` across 25 outer folds. Feature importance rank order was identical between the DNN and the logistic regression surrogate (Table 8), though absolute magnitude differences between explainer methods preclude direct quantitative comparison. Further strengthening of DNN SHAP attribution, including uncertainty quantification of `GradientExplainer` estimates, remains a future direction.

The feature penalty coefficient, $\lambda = 0.03$, was fixed prior to the sensitivity analysis and lies marginally below the identified stable plateau region ($\lambda = 0.04$–0.08). Examination of Fig. 5 indicates that the corresponding fitness value at $\lambda = 0.03$ (0.8426) is effectively unchanged relative to the plateau region, with negligible variation in AUC across this interval. Consequently, the practical impact on optimisation behaviour is limited. The resulting subset contained an average of 5.08 ± 0.27 features, remaining consistent with the intended sparsity range. Although $\lambda = 0.03$ imposes a slightly weaker sparsity constraint than values within the stable plateau, the penalty term remains active and continues to favour compact feature subsets during optimisation. A more principled optimisation strategy would involve systematic cross-validated tuning through a nested grid-search procedure, which is identified as an important direction for future refinement.

The per-split maximization over regularization strength C in (2) yields an optimistic fitness signal during optimization. Quantification of the resulting bias relative to a fixed-C evaluator remains an open item.

Adaptive weight and decay mechanisms showed no detectable effect on this dataset, as population diversity remained above the activation threshold throughout all runs. Their potential benefit in more heterogeneous or higher-dimensional search landscapes is not captured by the current experimental scope.

5.7 Future directions

External validation on contemporary and multi-site cardiovascular datasets, including MIMIC-IV, UK Biobank, and the Framingham Heart Study, is the most immediate priority. Temporal validation using time-based splits or prospective cohorts would additionally assess performance stability under real-world clinical workflow changes not captured by random cross-validation.

Future work should also investigate adaptive diversity thresholds that activate the full range of weight adaptation on datasets with different convergence characteristics, particularly in higher-dimensional or multi-modal search spaces where the current mechanism's dormancy would be resolved.

Further strengthening SHAP attribution alignment, including cross-fold averaging of interaction effects and uncertainty quantification of GradientExplainer estimates, remains a direction for future refinement. Finally, extending the

framework toward multi-objective FS incorporating feature acquisition cost, availability constraints, or fairness metrics may improve applicability in real-world clinical decision-support deployment.

Appendix A: Supplementary: mathematical notation summary

Table 12 Notation summary for Section 3.

Symbol	Definition
Dataset	
$\mathcal{D}$	Dataset of n labelled instances
$\mathbf{x}_i \in \mathbb{R}^d$	Feature vector; $d = 13$ (Core), $d = 10$ (Gen)
$y_i \in \{0, 1\}$	Disease label
FS objective	
$\mathbf{s} \in \{0, 1\}^d$	Binary feature mask
$\lvert\mathbf{s}\rvert_0$	Number of selected features
$f(\mathbf{s})$	Fitness: mean per-split best AUC (2)
$\lambda = 0.03$	Quadratic parsimony penalty
$\psi(\mathbf{s})$	Bonus: 0.015 if $5 \le \lvert\mathbf{s}\rvert_0 \le 7$, else 0
Optimization	
$N = 30$	Population size
$T_{\max} = 150$	Maximum iterations
$\tau = t/T_{\max}$	Normalized progress
$\mathbf{s}_i^{(t)}$	i-th continuous solution at iteration t
$\mathbf{s}^*$	Global best feature mask
$D(t)$	Population diversity (mean normalized Hamming)
$\omega(t)$	Adaptive hybrid weight; $\omega(0) = 0.5$
$\delta = 0.70$	EMA decay for ω
$\eta(t) = 0.15(1 - \tau) + 0.02$	Adaptive learning rate
$\beta(t) = 2e^{r(1-\tau)} \sin(2\pi r)$	WaOA spiral coefficient
$\alpha(t) = 2r\sqrt{-\log(r + 10^{-7})}$	Heavy-tailed perturbation coefficient
$C_{\text{stag}}(t)$	Stagnation counter
$L_{\text{stag}} = 15$	Stagnation threshold
$p_m(t, D)$	Diversity-adaptive mutation rate (10)
$T(t) = 0.1(1 - \tau)$	Simulated annealing temperature
$\varepsilon = 10^{-6}$	Logit clipping tolerance
Neural network	
$\boldsymbol{\Theta}$	All DNN parameters
$\lambda_{\text{L2}} = 0.001$	L2 weight decay
$\eta_0 = 0.001$	Initial Adam learning rate
Evaluation	
$K = 5$	MC cross-validation repeats
$\mathcal{C} = \{0.1, 1.0, 10.0\}$	LR regularization grid
$\phi_j(\mathbf{x})$	SHAP value for feature j
ECE	Expected Calibration Error (17)

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Data Availability The datasets used in this study were originally obtained from the UCI ML Repository,

Code Availability The source code for the proposed framework will be made available upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

Ethics Declaration Not applicable.

Code Availability The source code for the proposed framework will be made available upon reasonable request.

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